

COURSE TITLE : ADVANCED PROCESS CONTROL
COURSE CODE : 6211
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Process Control Strategies	18
2	Computerised process Control	19
3	PLC and SCADA	19
4	Modern Control Strategies	19
TOTAL		75

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the Batch and continuous process
	2	To comprehend the Basic process Control Strategies
2	1	To understand the Digital Signal Conditioning
	2	To understand the Use of Computers in Process Control
3	1	To understand the Programmable Logic Controller (PLC)
	2	To comprehend the Supervisory Control And Data Acquisition System (SCADA)
4	1	To understand Intelligent control
	2	To understand Virtual Instrumentation
	3	To know Fuzzy Logic System
	4	To understand P and I Diagram
	5	To know Controller Tuning

On completion of the study the student will be able

MODULE I PROCESS CONTROL STRATEGIES

1.1.0 To understand the Batch & continuous process

1.1.1 To explain the characteristics of Batch process with simple example

1.1.2 To describe the characteristics of Continuous Processes with simple example.

1.1.3 To Compare Batch and Continuous Process

1.2.0 To comprehend the Basic process Control Strategies

- 1.2.1 To explain single variable process control with example
- 1.2.2 To Illustrate independent variable process control with example
- 1.2.3 To describe interactive variable process control with example
- 1.2.4 To explain compound variable process control with example
- 1.2.5 To illustrate multivariable process control with example
- 1.2.6 To describe Feedback control system with suitable example
- 1.2.7 To describe Feed forward control system with suitable example
- 1.2.8 To compare Feedback control system & Feed forward control system
- 1.2.9 To illustrate Cascade control system with suitable example
- 1.2.10 To describe Ratio control system with suitable example
- 1.2.11 To explain Adaptive control system with suitable example.
- 1.2.12 To illustrate the Split Range Control system with suitable example

MODULE II COMPUTERISED PROCESS CONTROL

2.1.0 To understand Signal Conditioning

- 2.1.1 To define signal conditioning
- 2.1.2 To explain use of comparators & Universal Gates in process control
- 2.1.3 To illustrate the role of Alarms in process control
- 2.1.4 To describe the Functions of an Alarm system
- 2.1.5 To implement an Alarm Annunciating circuit using Digital gates & Comparators

2.2.0 To understand the Use of Computers in Process Control

- 2.2.1 To explain Data Loggers with suitable Block Diagram
- 2.2.2 To illustrate Data Acquisition System with suitable Block Diagram
- 2.2.3 To describe Supervisory control with suitable Block Diagram.
- 2.2.4 To explain Direct Digital Control with suitable Block Diagram
- 2.2.5 To compare Supervisory control and Direct Digital Control
- 2.2.6 To describe the block diagram and features of Centralized Computer control
- 2.2.7 To explain Distributed Control system (DCS) with its architecture
- 2.2.8 To describe general features of DCS.
- 2.2.9 To List advantage & disadvantage of DCS.

MODULE III PLC AND SCADA

3.1.0 To understand the Programmable Logic Controller (PLC)

- 3.1.1 To describe Relay Logic Control of Process with example
- 3.1.2 To explain the use of PLC in Industrial automation & its advantages
- 3.1.3 To illustrate basic concept of PLC
- 3.1.4 To explain the Block diagram of PLC
- 3.1.5 To explain the operation of PLC & its Connectivity
- 3.1.6 To describe input Module, Output Module and Memory types.
- 3.1.7 To illustrate PLC scan time
- 3.1.8 To describe different modes of operation of PLC
- 3.1.9 To describe selection criteria of PLC
- 3.1.10 To illustrate various instructions used in PLC
- 3.1.11 To explain the different methods of programming a PLC
- 3.1.12 To describe ladder diagram programming
- 3.1.13 To Write simple ladder diagram programming

3.2.0 To comprehend the Supervisory Control And Data Acquisition System (SCADA)

- 3.2.1 To explain SCADA with block diagram
- 3.2.2 To describe SCADA system components (Remote Terminal Unit, Master Station & Communication Infrastructure)
- 3.2.3 To describe general features of SCADA system

MODULE IV MODERN CONTROL STRATEGIES

4.1.0 To understand Intelligent Control

- 4.1.1 To describe Intelligent Control
- 4.1.2 To explain the features of Intelligent control

4.2.0 To understand Virtual Instrumentation

- 4.2.1 To describe Virtual Instrumentation
- 4.2.2 To illustrate the Virtual Instrument and Traditional Instrument with figure
- 4.2.3 To describe the role of hardware and software in Virtual Instrumentation
- 4.2.4 To explain the Virtual Instrumentation for Test,Control and Design
- 4.2.5 To know the graphical system design using LabVIEW
- 4.2.6 To compare Text based programming and graphical programming
- 4.2.7 To List the advantages of LabVIEW.

4.3.0 To know Fuzzy Logic System

- 4.3.1 To describe Fuzzy Logic System
- 4.3.2 To illustrate the block diagram of Fuzzy Controller
- 4.3.3 To explain Fuzzyfier and Defuzzyfier

4.4.0 To understand P and I Diagram

- 4.4.1 To describe P & I Diagram
- 4.4.2 To explain graphical elements ,line symbols and Instrument Identification

4.5.0 To know Controller Tuning

- 4.5.1 To Describe openloop method of controller tuning
- 4.5.2 To illustrate Ziegler-Nichols methods of Controller tuning

CONTENT DETAILS

MODULE I

Characteristics of Batch process & Continuous Processes – Comparison - Single variable, Independent variable, Interactive variable - compound variable - multivariable control -Feedback control system - Feed forward control system – Comparison - Cascade control system - Ratio control system - Adaptive control system - Split Range Control system

MODULE II

signal Conditioning – Block diagram of digital signal Conditioning -comparators & Universal Gates - role of Alarms - Functions of an Alarm system - Alarm Annunciating circuit -Data Loggers with block diagram - Data Acquisition System with suitable Block Diagram-Supervisory control with block diagram-Direct Digital Control with block diagram –Comparison - Centralized Computer control – features –block diagram - DCS with its architecture - general features-advantages - disadvantages

MODULE III

Relay Logic Control of Process - PLC in Industrial automation -basic concept -Block diagram of PLC - operation – Connectivity - input Module, Output Module and Memory types - PLC scan time - modes of operation of PLC - selection criteria - instructions set - methods of programming a PLC - ladder diagram programming-simple program - SCADA with block diagram - general features- system components

MODULE IV

Intelligent Control - Features - Virtual Instrumentation - Virtual instrument - traditional Instrument-Hardware and software-Test control and design-LabVIEW - Text based programming and graphical programming – advantages - Fuzzy Logic System – Block Diagram-Fuzzyfier-defuzzyfier - P & I Diagram - Graphical element-line symbol-Instrument identification - Controller tuning - Open loop method-Ziegler Nichols Method.

REFERENCES

1. Curtis .D.Johnson , Process control instrumentation technology , PHI
2. George stephanopoulos, Chemical Process Control- PHI
3. Krishnakanth , Computer based Industrial Control- PHI
4. John webb , Programmable Logic Controllers - Macmillan Publishing
5. P.Sai Krishna , Process Control Engineering , IK
6. Considine , Applications of Computers in Process Control
7. Liptak , Handbook of Process Control
8. Jovitha Jerome,Virtual Instrumentation using LabVIEW,PHI